Assignment 01: Foundations of Computational Learning

Case Study Analysis: Stochastic Systems & Linear Approximation

Module: Introduction to Machine Learning

Instructions

- **Type:** Essay and Procedural Analysis.
- **Goal:** This assignment evaluates your ability to translate a complex real-world problem into a mathematical machine learning framework. You are expected to demonstrate theoretical understanding, mathematical formalism, and procedural application.
- **Format:** Answers must be cohesive, using proper mathematical notation $(\theta, \mathbf{X}, y)$ as defined in the lecture notes.

Case Study: Project Heliopolis (Solar Energy Prediction)

You have been hired as a lead Data Scientist for *Heliopolis Energy*, a company managing a massive solar farm in a desert region. The engineering team is currently using a **Classical Deterministic Algorithm** to predict the daily energy output (y, in Megawatts) based on sunlight duration (x_1).
However, their predictions are consistently failing. Even on days with identical sunlight duration, the actual energy output fluctuates wildly due to unmeasured factors like dust accumulation on panels, ambient temperature, and cloud density.
Your task is to transition the company's prediction system from a **Deterministic Rule-Based System** to a **Stochastic Machine Learning Model**.

Part 1: Abstraction and Formal Representation (Cognitive Analysis)

The engineering team provides you with raw logs containing unstructured data: *"Day 1, 10 hours sun, 40 degrees C, produced 50MW"*.

Q1. Data Vectorization: Construct the mathematical definition of the Dataset $\mathcal{D}$ for this problem.

- Define the Feature Vector $\mathbf{x}^{(i)}$ including at least three relevant features you would extract from the environment.
- Define the dimensions of the Feature Matrix $\mathbf{X}$ and Label Vector $\mathbf{Y}$ if you collect data for $N = 365$ days.

Q2. The Hypothesis Space: Propose a Linear Hypothesis model $h_\theta(\mathbf{x})$ for this problem. Explicitly interpret the meaning of the parameter vector θ in the context of solar energy. What does a high positive value for a specific weight w_j imply about its corresponding feature?

Part 2: The Stochastic Argument (Theoretical Foundation)

The Chief Engineer argues: *"Physics is precise. If we know the photon count, we should calculate energy exactly. Why do we need probability?"*

Q3. Refuting Determinism: Write a cohesive argument explaining why an analytical solution $y = f(x)$ is impossible in this environment. Introduce the concept of **Aleatoric Uncertainty** (noise) and formally express the relationship between inputs and outputs using the stochastic formulation $y = f(x) + \epsilon$.

Part 3: The Optimization Cycle (Procedural Application)

This section evaluates your psychomotor/procedural understanding of the training loop.

Imagine the model is currently in its initial state with random parameters. We receive a single data point:

$$\mathbf{x}^{(1)} = [2, 1]^T \quad \text{(Features)}$$

$$y^{(1)} = 5 \quad \text{(Actual Energy Output)}$$

Current Parameters: $\theta = \{\mathbf{w} = [1.5, 1.0]^T, b = 0\}$

Q4. Manual Forward Pass: Calculate the model's prediction $\hat{y}^{(1)}$ using the current parameters. Show your step-by-step matrix algebra.

Q5. Loss Calculation: Compute the Squared Error for this specific sample. Then, explain in words: what does this scalar value communicate to the Optimizer?

Q6. Gradient Intuition: Without performing the full derivative calculus, describe the **mechanism** of the update step. If the prediction is too high (overestimation), conceptually, how must the parameters θ change? Sketch a simple diagram showing the "descent" on a loss surface.

Part 4: Evaluation and Generalization (HOTS)

After training for 1,000 iterations, the model achieves a Training Loss (MSE) of **0.001** (near zero). The engineers are celebrating, claiming the model is perfect.

Q7. The Overfitting Trap: You run the model on data from a neighboring region (Validation Set) and get an MSE of **50.0**.

- Diagnose this phenomenon. What has the model likely "learned" instead of the physics of solar energy?
- Explain mathematically why $J_{train}(\theta) \approx 0$ does not imply $J_{val}(\theta) \approx 0$.